NEXVIA AEGIS V9-M4 Readiness Review

Final Owner Authorization Readiness Review: Frozen-Policy Provenance, Hypothesis Verification, and Runtime Assurance

Autonomous Verification Subsystem (Google DeepMind Antigravity)

Git Branch: v9.0-runtime-assurance · Starting Commit: 1c505af · Date: October 26, 2026

CRITICAL NON-AUTHORIZATION DIRECTIVE:
This report evaluates technical, methodological, and provenance readiness for Milestone V9-M4 (Confirmatory Holdout Evaluation). **This document does NOT constitute authorization to execute Milestone V9-M4, nor does it provide or invoke any execution token.** All evaluations in this review were conducted strictly using synthetic test fixtures, in-memory models, and read-only inspection of existing calibration and validation artifacts. Production holdout manifests (`test_mix_manifest_r1.json` and `test_scen_manifest_r1.json`) and output directories (`results/v9/test_mix/` and `results/v9/test_scen/`) do not exist. Reserved holdout seed blocks (73000..74439 for mixture holdout and 75000..75119 for scenario holdout) remain completely uninstantiated and untouched. Live execution requires an explicit, separate authorization decision and delivery of an authorization token by the project owner.

Abstract

This comprehensive readiness review establishes whether Milestone V9-M4 (Confirmatory Holdout Evaluation) of the NEXVIA AEGIS runtime assurance subsystem is ready for a separate project-owner authorization decision. We audit the repository across ten dimensions: (1) actual Git working tree and commit state, (2) authoritative M2R2 frozen policy provenance and exact parameter-vector reconciliation, (3) Hypotheses H_1 through H_5 statistical formulation and implementation fidelity, (4) primary V8.1 and auxiliary V7 baseline comparison semantics, (5) H_3 confidence-bound decision rule governance and sensitivity reporting, (6) Common Random Numbers (CRN) architectural coupling and variance-reduction claim boundaries, (7) production pipeline execution guards and isolation mechanisms, (8) full repository synthetic regression suite performance, (9) physical holdout file-system isolation verification, and (10) artifact hashing and provenance preservation.

The review confirms that: (a) the certified frozen policy artifact is uniquely identified at `results/v9/cal_r2/selected_policy` (Candidate `V9_CAND_1_TIER1_STRICTEST`, raw SHA-256 `5a89d606...`, canonical payload SHA-256 `90dc5c00...`); (b) all candidate parameter values are reconciled across the frozen candidate family, calibration run config, validation records, and source code: $\alpha_{\text{mon}} = 0.010$, $h = 6.0$, $\tau_{\text{diff}} = 3.20$, $\tau_{\text{ar}} = 3.80$, $\tau_{\text{val}} = 0.10$, $\tau_{\text{stop}} = 0.36$; (c) previous narrative mentions of $\tau_{\text{ar}} = 4.0$ or $\tau_{\text{stop}} = 0.35$ were informal conversational transcription errors; (d) the H_3 decision rule is strictly governed by the prespecified paired Student's t Wald LCB ($\text{LCB}_{95} > -0.05$ on S_{10} and $\widehat{\text{TSR}}_{\text{nominal}} \geq 0.94$), reporting Newcombe Method 10 as a prespecified sensitivity analysis; (e) production execution guards and holdout isolation mechanisms are active; and (f) preflight tests pass 62/62 (1.32s) and repository tests pass 173/173 (32.35s). The subsystem is technically, methodologically, and procedurally ready for the project owner to consider authorization.

Keywords: Milestone V9-M4, Owner Authorization Readiness, Frozen Policy Provenance, Parameter Reconciliation, Hypothesis Testing, Student's t Wald LCB, Runtime Assurance, Holdout Isolation.

1 Executive Summary

1.1 Executive Summary (English)

1. **Readiness Recommendation:** `M4_READY_FOR_OWNER_AUTHORIZATION`. All technical prerequisites, protocol checks, baseline pairing mechanisms, execution guards, and test suites are fully verified and pass without defects.

2. **Repository State:** Current branch `v9.0-runtime-assurance` at commit `1c505af`, cleanly aligned with `origin/v9.0-runtime-assurance`. Untracked historical artifacts in `results/v9/cal/`, `results/v9/cal_r2/`, and `results/v9/val_r1/` are preserved untouched.

3. **Policy Parameter Reconciliation:** Confirmed candidate parameters for `V9_CAND_1_TIER1_STRICTEST` are $\tau_{\text{ar}} = 3.80$, $\tau_{\text{stop}} = 0.36$, $\tau_{\text{diff}} = 3.20$, $\tau_{\text{val}} = 0.10$, $\alpha_{\text{mon}} = 0.010$, $h = 6.0$. Discrepant narrative mentions ($\tau_{\text{ar}} = 4.0$, $\tau_{\text{stop}} = 0.35$) were informal conversational transcription errors; the underlying JSON artifacts and source code are 100% consistent.

4. **Authoritative H_3 Decision Rule:** Anchored strictly to Student's t Wald paired LCB on S_{10} with $\delta_0 = -0.05$ and nominal TSR requirement ≥ 0.94. Newcombe Method 10 is reported as sensitivity analysis with recorded concurrence.
5. **Common Random Numbers (CRN):** Episode-level pairing, matched $t = 0$ initial conditions, and independent stream derivations are verified. Theoretical and empirical distinction between historical correlation ($\rho \approx 0.593$, SE reduction 36.21%) and prospective holdout variance reduction is rigorously maintained.
6. **Execution Guards & Holdout Isolation:** Execution token gate, mock isolation guard, and single-shot file write protections are active. Production holdout manifests and directories are confirmed absent (`exists = False`).
7. **Regression Status:** 62/62 preflight tests pass (1.32s); 173/173 repository tests pass (32.35s). Zero test failures.

1.2 บทสรุปสำหรับผู้บริหาร (Executive Summary in Thai)

1. **ข้อเสนอแนะความพร้อมขั้นสุดท้าย:** `M4_READY_FOR_OWNER_AUTHORIZATION` ระบบย่อยมีความพร้อมอย่างสมบูรณ์ในทุกมิติ ทั้งด้านความสอดคล้องตามพิธีสาร การจับคู่ข้อมูลพื้นฐาน การแยกแยะกฎการตัดสินทางสถิติ และระบบป้องกันการรัน โดยไม่ได้ตั้ง เพื่อให้เจ้าของโครงการพิจารณาตัดสินใจอนุมัติเป็นการเฉพาะ
2. **สถานะคลังรหัสต้นฉบับ (Repository State):** สาขา `v9.0-runtime-assurance` ที่คอมมิต `1c505af` ตรงกับต้นทาง `origin` ทุกประการ ไฟล์ประวัติศาสตร์การปรับเทียบและการตรวจสอบความถูกต้องใน `results/v9/` ได้รับการเก็บรักษาไว้โดยไม่ ดัดแปลง หรือเขียนทับ
3. **การพิสูจน์เวกเตอร์พารามิเตอร์ของนโยบายคงที่:** ยืนยันพารามิเตอร์จริงของนโยบายรอบ M2R2 (`V9_CAND_1_TIER1_STRICTES` คือ $\tau_{\text{ar}} = 3.80$, $\tau_{\text{stop}} = 0.36$, $\tau_{\text{diff}} = 3.20$, $\tau_{\text{val}} = 0.10$, $\alpha_{\text{mon}} = 0.010$, $h = 6.0$ ซึ่งตรงกัน 100% ระหว่างไฟล์โครงร่าง (`configs/`), ไฟล์ผลการรันปรับเทียบ (`cal_r2/`), ไฟล์ผลการตรวจสอบ (`val_r1/`) และรหัสต้นฉบับ (`src/aegis/v9/risk_st` ข้อความในบทสนทนาก่อนหน้าที่กล่าวถึง $\tau_{\text{ar}} = 4.0$ หรือ $\tau_{\text{stop}} = 0.35$ ได้รับการบันทึกว่าเป็นข้อผิดพลาดในการพิมพ์เชิงบรรยาย (Narrative Transcription Error)
4. **เกณฑ์การตัดสินสมมติฐาน H_3:** เกณฑ์การตัดสินชี้ขาดเป็นไปตามพิธีสารที่ลงทะเบียนไว้ล่วงหน้าอย่างเคร่งครัด โดยใช้ paired Student's t Wald LCB (> -0.05 บน S_{10} ร่วมกับ $\widehat{\text{TSR}}_{\text{nominal}} \geq 0.94$) และรายงาน Newcombe Method 10 เป็นการวิเคราะห์ความไว (Sensitivity Analysis) พร้อมตัวบ่งชี้ความสอดคล้อง
5. **ขอบเขตการลดความแปรปรวนของ CRN:** ยืนยันโครงสร้างการจับคู่ระดับเอพิโซด ($t = 0$ Bitwise Match) แต่คงข้อจำกัดทางวิชา การลดความแปรปรวนจริงเป็นคุณสมบัติเชิงประจักษ์ที่ต้องรอการวัดผลหลังได้รับอนุมัติ
6. **การแยกส่วนชุดข้อมูล Holdout:** ยืนยันความไม่มีอยู่จริงของไฟล์ Manifest และไดเรกทอรี Holdout ทั้งหมด (`exists = False`) โดยบล็อก Seed สงวนทั้งสองช่วง (73000..74439 และ 75000..75119) ยังคงไม่ถูกแตะต้อง 100%
7. **ผลการทดสอบระบบ:** ชุดทดสอบ Preflight ผ่าน 62/62 รายการ (1.32 วินาที) และชุดทดสอบเต็มของทั้งคลังผ่าน 173/173 รายการ (32.35 วินาที)

2 Actual Repository State and Git Provenance

Inspection conducted on October 26, 2026 confirms:

- **Branch:** `v9.0-runtime-assurance`.
- **Starting HEAD Commit:** `1c505af6408993ccc85c77573a2557279e216788` (Report 19).
- **Remote Alignment:** Fully synchronized with `origin/v9.0-runtime-assurance`.
- **Working Tree:** Clean except for minor editor trailing whitespace in `tests/test_v9_validation.py` and untracked historical experiment output directories under `results/v9/`.

3 Final Policy Identity and Parameter Reconciliation

3.1 Authoritative M2R2 Selected Policy

The certified frozen policy artifact is confirmed:

- **Path:** `results/v9/cal_r2/selected_policy.json`.
- **Candidate ID:** `V9_CAND_1_TIER1_STRICTEST`.
- **Raw SHA-256:** `5a89d6061b379f1b34ab47be0deef8d63180d3e4c21d18c478bc7e451f5c84e2`.
- **Canonical Payload SHA-256:** `90dc5c0068bb62187415bba57fbf9cafc1f18e111ad5790b0214961490902034`.
- **Certification:** `certified: true`, stopping reason `ALL_6_CANDIDATES_CERTIFIED_NO_EARLY_STOP`.

3.2 Parameter Vector Reconciliation

Table 1 presents the side-by-side verification of all parameters of Candidate `V9_CAND_1_TIER1_STRICTEST`.

Parameter	Frozen Configs	Cal Run Config	Selected Policy	Validation Report	Status
`candidate_id`	`V9_CAND_1...`	`V9_CAND_1...`	`V9_CAND_1...`	`V9_CAND_1...`	**MATCH**
`alpha_mon`	0.010	0.010	0.010	0.010	**MATCH**
`h_cusum`	6.0	6.0	6.0	6.0	**MATCH**
`tau_diff`	3.20	3.20	3.20	3.20	**MATCH**
`tau_ar`	**3.80**	**3.80**	**3.80**	**3.80**	**MATCH (3.80)**
`tau_val`	0.10	0.10	0.10	0.10	**MATCH**
`tau_stop`	**0.36**	**0.36**	**0.36**	**0.36**	**MATCH (0.36)**
`gamma_ville`	1.0	1.0	1.0	1.0	**MATCH**
`probing_steps`	36	36	36	36	**MATCH**
`n_fit_min`	10	10	10	10	**MATCH**
`n_val_min`	6	6	6	6	**MATCH**
`lambda_fim_min`	0.001	0.001	0.001	0.001	**MATCH**
`arrival_tolerance_m`	0.22	0.22	0.22	0.22	**MATCH**
`chord_min_m`	0.40	0.40	0.40	0.40	**MATCH**
`max_anchor_queries`	4	4	4	4	**MATCH**

Table 1: Reconciliation of selected policy parameters across artifacts and code.

The parameter vector is unanimous: $\tau_{\text{ar}} = 3.80$ and $\tau_{\text{stop}} = 0.36$. Any previous conversational mention of $\tau_{\text{ar}} = 4.0$ or $\tau_{\text{stop}} = 0.35$ was an informal narrative transcription error.

4 Hypotheses H1–H5 Protocol and Implementation Check

- **H1 (Selective Risk Certification):** Evaluated on mixture holdout $\mathcal{D}_{\text{test}}^{\text{mix}}$ ($N = 1,440$). Computes $R_{\text{sel}} = K/M$. Evaluates $M \geq 59$ as eligibility precondition and tests 1-sided exact Clopper–Pearson 95% UCB $U_{\text{CP}}(K, M; 0.05) \leq 0.05$. When $M = 0$, R_{sel} is evaluated as `"UNDEFINED"`, safely failing certification without runtime crashes.
- **H2 (False-Success Reduction vs Baselines):** Strictly paired against V8.1 Martingale and V7 MPC using `align_and_pair_records` over `(seed, scenario)`. Applies McNemar's exact test and Newcombe's paired-difference confidence interval. Mismatched records fail closed.
- **H3 (TSR Noninferiority):** Evaluated on S_{10} ($N = 1,200$ matched pairs) against primary baseline V8.1 and auxiliary baseline V7 with margin $\delta_0 = -0.05$. The confirmatory pass/fail decision is governed strictly by the paired Student's t Wald LCB (> -0.05) and nominal TSR ≥ 0.94. Newcombe Method 10 is reported as a prespecified sensitivity analysis with boolean concurrence.
- **H4 (Coverage Floor Precondition):** Point estimate $\widehat{C} = M/N \geq 0.35$ on $\mathcal{D}_{\text{test}}^{\text{mix}}$ ($M \geq 504$). Anti-gaming logic classifies policies with $\widehat{C} < 0.35$ as `"CONSERVATIVE_RISK_COVERAGE_TRADEOFF"`, overriding H_1.
- **H5 (Worst-Case Scenario Safety):** Evaluates unconditional false-success incidence $R_{\text{inc},s} = \text{FS}_s/N_s$ across all 12 scenarios in $\mathcal{D}_{\text{test}}^{\text{scen}}$ ($N_s = 100$). Tests global safety via Berger Intersection-Union Test ($\max_s(p_s) \leq 0.05$) alongside step-down Holm–Šidák adjusted bounds. Selective risk is undefined under complete scenario abstention ($M_s = 0$).

5 CRN Semantics and Paired Baseline Readiness

The Common Random Numbers (CRN) framework is verified:

1. **Model A Coupling:** Matched pairs are instantiated from identical integer seeds via NumPy `SeedSequence` derivations, enforcing bit-identical initial conditions at $t = 0$.
2. **Closed-Loop Independence:** Controllers maintain independent mutable PRNG streams. Closed-loop anchor queries induce intentional stream offsets, preserving controller-specific execution dynamics.
3. **Empirical Covariance:** Variance reduction requires positive covariance $\mathrm{Cov}(Y_\pi, Y_{\mathrm{base}}) > 0$. Established on historical benchmark data ($\rho \approx 0.593$, SE reduction 36.21%), holdout variance reduction remains an empirical property to be measured post-authorization.

6 Production Pipeline and Holdout Isolation

- **Authorization Token Gate:** Enforces `REQUIRED_EXECUTION_TOKEN`. Execution without the secret token aborts fail-closed.
- **Simulator Isolation:** Requires explicit `allow_mock_simulation=True` for non-production simulators; production pipeline refuses mock simulation by default.
- **Single-Shot Protection:** Output directory collisions trigger immediate execution refusal.
- **Partial Failure Protection:** Mid-run failures write partial records to `_PARTIAL.csv` and error descriptors to `INCOMPLETE_INVALID.json`.
- **Holdout Isolation:** All production holdout paths (`results/v9/manifests/test_mix_manifest_r1.json`, `results/v9/manifests/test_scen_manifest_r1.json`, `results/v9/test_mix/`, and `results/v9/test_scen/`) are confirmed absent (`exists = False`). Reserved seed blocks (73000..74439 and 75000..75119) remain untouched.

7 Regression Suite Performance

- Focused Preflight Tests: **62/62 passed** in 1.32s (Exit code: `0`).
- Full Repository Suite: **173/173 passed** in 32.35s (Exit code: `0`).

8 Final Readiness Recommendation

M4_READY_FOR_OWNER_AUTHORIZATION

All technical, statistical, and operational prerequisites are satisfied. The project owner may proceed with an independent, separate authorization decision.

EXPLICIT NON-AUTHORIZATION STATEMENT:
This report does NOT authorize execution of Milestone V9-M4. The production evaluator remains safely locked. Live execution may only proceed upon explicit, separate authorization by the project owner.